

Practicum 2: Learning-Rate Sensitivity of the Perceptron Learning Algorithm

Objective

Repeat the four tasks of Practicum 1 across a grid of learning rates and establish, both empirically and analytically, what the learning rate α does and does not control in the Perceptron Learning Algorithm (PLA). You will track how the number of updates, the number of epochs to convergence, and the geometry of the final decision boundary respond to α , and you will explain the cases in which α has no effect at all.

The Learning-Rate Grid

Use the same error-driven update rule as Practicum 1, with the Heaviside prediction $\hat{y} = 1$ if $z \geq 0$ and $\hat{y} = 0$ otherwise:

- Weight Update: $w_j := w_j + \alpha (y - \hat{y}) x_{ij}$
- Bias Update: $b := b + \alpha (y - \hat{y})$

Run every task once for each of the eleven learning rates

$$\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}.$$

The case $\alpha = 0.0$ is deliberately included; you are expected to run it, observe what happens, and explain it in Task 5.

Fixed initialisation and presentation order

Because the effect of α is only visible relative to the initial parameters (see Task 5), the initialisation is prescribed and must be identical for every learning rate and every gate:

$$w_1 = 1, \quad w_2 = -1, \quad b = 0.5.$$

Present the truth-table rows in the fixed order $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$ in every epoch. Do not shuffle. Do not use random initialisation anywhere in this practicum; the comparison across α is meaningless unless every run starts from the same point and sees the same data sequence.

Assignment Tasks

Task 1: The AND Gate

- For each α in the grid, initialise as prescribed and execute the PLA on the 2-input AND truth table until convergence (zero errors in a full epoch), subject to a safety cap of 200 epochs.
- Record, for each α : the epoch of convergence, the total number of weight updates performed, and the final parameters (w_1, w_2, b) .

Task 2: The OR Gate

- Repeat Task 1 for the 2-input OR truth table.
- Record the same three quantities for each α .

Task 3: The XOR Gate

- For each α , run the PLA on the 2-input XOR truth table for exactly 10 epochs.
- Record the total number of updates, and the parameter vector (w_1, w_2, b) at the end of every epoch.
- Identify whether the parameters enter a cycle. If so, state the period of the cycle (in updates) and the set of parameter vectors visited. Determine whether the period depends on α .

Task 4: The Half Adder

- Train the Carry perceptron and the Sum perceptron simultaneously, each from the prescribed initialisation, for a maximum of 10 epochs, for each α .
- For each α , document which perceptron converges, at which epoch, and which fails.

Task 5: Analysis

Answer the following in full sentences, with derivations where a derivation is asked for.

- The degenerate rate.** State precisely what happened for $\alpha = 0.0$ in every task and explain why, directly from the update rule.
- Scale invariance.** Suppose the initialisation had instead been $w_1 = w_2 = b = 0$. Prove that, for any two learning rates $\alpha, \alpha' > 0$, the two runs make exactly the same sequence of predictions, perform updates on exactly the same rows, and produce decision boundaries that coincide at every step. (Hint: show by induction that the parameters after k updates satisfy $(w^{(k)}, b^{(k)}) = \alpha (u^{(k)}, c^{(k)})$ for a sequence $(u^{(k)}, c^{(k)})$ that does not depend on α , and observe what the Heaviside function does to positive scaling.)
- Rate–initialisation equivalence.** Show that running the PLA with learning rate $\alpha > 0$ from initialisation $(w^{(0)}, b^{(0)})$ produces the same sequence of decision boundaries as running it with learning rate 1 from initialisation $(w^{(0)}/\alpha, b^{(0)}/\alpha)$. Conclude that, for the perceptron, the learning rate is equivalent to a choice of scale for the initial parameters and nothing more.
- Reconciling (b) with your data.** Your convergence tables from Tasks 1 and 2 should show that the epoch of convergence *does* vary with α under the prescribed nonzero initialisation. Use (c) to explain why this is consistent with (b). Then explain why the dependence on α is not necessarily monotone.
- Geometry of the final boundary.** For each converged run in Tasks 1 and 2, compute the signed distance from every input point to the final hyperplane, $d_i = (w^\top x_i + b)/\|w\|_2$, and the margin $\min_i |d_i|$. Tabulate the margin against α and comment on whether a smaller learning rate produces a “better” separator in any geometric sense.

Required Deliverables for All Tasks

1. Convergence Summary Tables

For each of Tasks 1–4, one table with a row per learning rate ($\alpha = 0.0, 0.1, \dots, 1.0$) and columns for: epoch of convergence (or “no convergence”), total updates, final (w_1, w_2, b) , and, for Tasks 1–2, the margin from Task 5(e).

2. Step-by-Step Execution Tables

As in Practicum 1, construct a tabular trace for every row of the truth table across every epoch, listing the input vector, target, net input $z = w_1x_1 + w_2x_2 + b$, prediction $\hat{y}$, the update applied, and the resulting parameters. Full traces are required for every α . Place the traces for $\alpha \in \{0.1, 0.5, 1.0\}$ in the main body of the report and the remaining traces in an appendix, clearly labelled by task and by α .

3. Hyperplane Plots

All plots are 2D Cartesian plots in the (x_1, x_2) plane with the four data points colour-coded by target class, the weight vector W drawn as an arrow, and the decision boundary $w_1x_1 + w_2x_2 + b = 0$ drawn as a line. Three families of plots are required.

- (a) **Per-update plots.** For each gate and each $\alpha > 0$, a plot at every step at which a weight update occurs, exactly as in Practicum 1. Arrange the plots for one (gate, α) pair as a single grid of small panels, one panel per update, so that the rotation of the boundary can be read left to right.
- (b) **Overlay plots.** For each of AND, OR, Carry and Sum: one figure showing the *final* hyperplane of every α in the grid on the same axes, drawn in a colour that varies with α , with a legend or colour bar. Where hyperplanes coincide, say so in the caption.
- (c) **Convergence curves.** For AND and OR: epoch of convergence against α , and total updates against α . For XOR: the trajectory of (w_1, w_2, b) over the update index for every α , on one figure with three subplots (one per parameter), so that the oscillation and its amplitude as a function of α are visible.

4. Written Analysis

The answers to Task 5, including the two proofs, typeset in full.

Submission Guidelines

- **Implementation:** Python with NumPy and Matplotlib only. The PLA must be implemented from scratch; no scikit-learn or equivalent library implementations of the perceptron are permitted. Your code must be included as an appendix and must reproduce every table and figure from a single run.
- **Document Format:** Compile all deliverables (summary tables, execution tables, plots, and analysis) into a single, well-organised PDF document.
- **File Naming Convention:** Name your final submission file strictly as RollXX_FirstName_Assign2.pdf, where XX is your class roll number; e.g. Roll01_Adam_Assign2.pdf for a person named Adam Smith having class roll 01.